Comparison of Lightweight Encryption Algorithms

Progress Report - Phase 1

Rajesh Kumar Jena (B122088) Soumil Kumar (B122114) Punit Kumar (B422041)

International Institute of Information Technology
Bhubaneswar
Supervisor: Prof. Debasish Jena

March 2026

Outline

1. Introduction
2. Algorithm Overview
3. Project Status
4. Methodology
5. Results and Discussion
6. Conclusion and Future Work

Executive Summary

Current Status:

- **Implementation**: COMPLETE - Core ciphers (PRESENT, SPECK, ASCON) implemented in Rust
- **Testing**: PARTIAL - Basic encryption/decryption tests pass, NIST KAT tests pending
- **Host Benchmarks**: PARTIAL - Initial metrics collected, additional analysis in progress
- **Pico Benchmarks**: PARTIAL - Initial benchmarks run, detailed analysis pending
- **ROM/RAM Analysis**: PARTIAL - Host analysis complete, Pico-specific analysis pending

Executive Summary: Key Findings

Areas for Improvement (Future Work):

- NIST KAT (Known Answer Test) vector verification
- Constant-time implementations for side-channel resistance
- Energy consumption analysis

Key Findings:

- SPECK is the fastest (32 Mbps on host, 24 Mbps on Pico), approximately 70x faster than PRESENT
- PRESENT has the smallest code footprint, designed for hardware efficiency
- ASCON provides built-in authenticated encryption with modern security (NIST Standard 2023)
- CTR mode offers best performance among encryption modes for SPECK

Motivation

- IoT deployments place cryptography on devices with limited CPU, memory, and energy
- Conventional algorithms are often too expensive for such platforms
- *Lightweight cryptography* targets security with lower computational and memory cost

Problem Statement and Objectives

Problem Statement

- AES, RSA, and ECC often exceed microcontroller resource limits
- Lightweight cryptography addresses this gap without sacrificing core security goals

Objectives

- Implementation of lightweight ciphers (PRESENT, SPECK, ASCON) in Rust
- Benchmarking on Raspberry Pi Pico and host machine
- Performance analysis and comparison
- Security considerations for embedded systems

Selected Algorithms

- **PRESENT**: Hardware-oriented 64-bit SPN cipher with 80/128-bit keys
- **SPECK**: Software-optimized ARX cipher (NSA-designed)
- **ASCON**: NIST Lightweight Cryptography Standard (2023) - authenticated encryption

PRESENT

- 64-bit SPN block cipher
- 80-bit and 128-bit keys
- Designed for ultra-lightweight hardware
- Round steps: AddRoundKey, S-box, permutation

```
const SBOX: [u8; 16] = [
    0xC, 0x5, 0x6, 0xB, 0x9, 0x0, 0xA, 0xD,
    0x3, 0xE, 0xF, 0x8, 0x4, 0x7, 0x1, 0x2,
];

fn sbox_layer(state: u64) -> u64 {
    let mut result = 0u64;
    for i in 0..16 {
        let nibble = ((state >> (i * 4)) & 0xF) as usize;
        result |= u64::from(SBOX[nibble]) << (i * 4);
    }
    result
}
```

SPECK

- Software-optimized ARX cipher
- Feistel-like word-based structure
- Efficient on microcontrollers

```rust
pub enum SpeckVariant {
    Speck32_64,
    Speck48_72,
    Speck48_96,
    Speck64_96,
    Speck64_128,
}

impl SpeckVariant {
    pub const fn rounds(self) -> usize {
        match self {
            Self::Speck64_96 => 26,
            Self::Speck64_128 => 27,
            _ => 22,
        }
    }
}
```

ASCON

- NIST Lightweight Cryptography Standard (2023)
- Sponge-duplex construction
- 320-bit internal state
- Supports authenticated encryption and hashing

```
const ROUNDS_A: usize = 12;
const ROUNDS_B: usize = 6;

const fn p_sbox(s0: u64, s1: u64, s2: u64, s3: u64, s4: u64)
    -> (u64, u64, u64, u64, u64) {
    let s0 = s0 ^ s4;
    let s4 = s4 ^ s3;
    let s2 = s2 ^ s1;
    let t0 = (!s0) & s1;
    let t1 = (!s1) & s2;
    (s0, s1, s2, s3, s4)
}
```

Project Status

Task	Status	Notes
Literature Survey	Complete	100%
Algorithm Implementation	Complete	Core ciphers done
Test Vector Verification	Partial	Basic tests pass, KAT pending
Host Benchmarks	Partial	Initial metrics collected
Pico Hardware Benchmarks	Partial	Initial benchmarks run
ROM/RAM Analysis	Partial	Host analysis done

Completed Tasks and Pending Tasks

Completed Tasks

- Rust implementations: PRESENT, SPECK, and ASCON
- Mode implementations: ECB, CBC, CTR for PRESENT and SPECK
- Embedded `no_std` library for target hardware
- Basic validation: test vectors, consistency checks, ASCON tag verification
- Host machine validation tests

Pending Tasks

- Final comparison and conclusions
- ROM/RAM memory analysis on target hardware
- Energy consumption estimation
- Final report compilation

Methodology

Evaluation Criteria

- **Execution Time**: ns per block (measured via hardware timer)
- **Cycles Per Byte (CPB)**: Platform-independent efficiency metric
- **Throughput**: Mbps for comparing speed
- **Memory**: ROM/RAM footprint analysis

Hardware Platform

- Raspberry Pi Pico (ARM Cortex-M0+, 133 MHz, 264 KB SRAM)
- Rust `no_std` embedded implementation
- Release mode with LTO and size optimization

Benchmarking Approach

- Hardware timer for cycle-accurate measurements
- 1000+ iterations for statistical accuracy
- Warm-up runs before measurements
- Test vector validation for correctness

Host Machine Benchmarks

Benchmarks on x86_64 validate the implementations; absolute values differ from Pico, but relative comparisons remain useful.

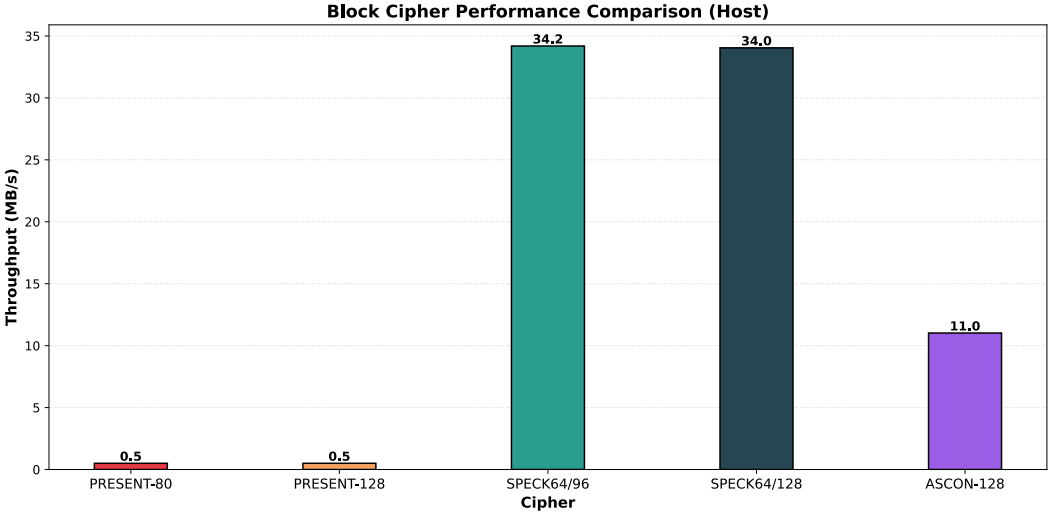
Algorithm	ns/block	CPB	Mbps
PRESENT-80	16,949	2,119	0.47
PRESENT-128	16,990	2,124	0.47
SPECK-64/96	246	31	32.52
SPECK-64/128	245	31	32.65
ASCON-128	1,518	95	10.54

Raspberry Pi Pico Benchmarks

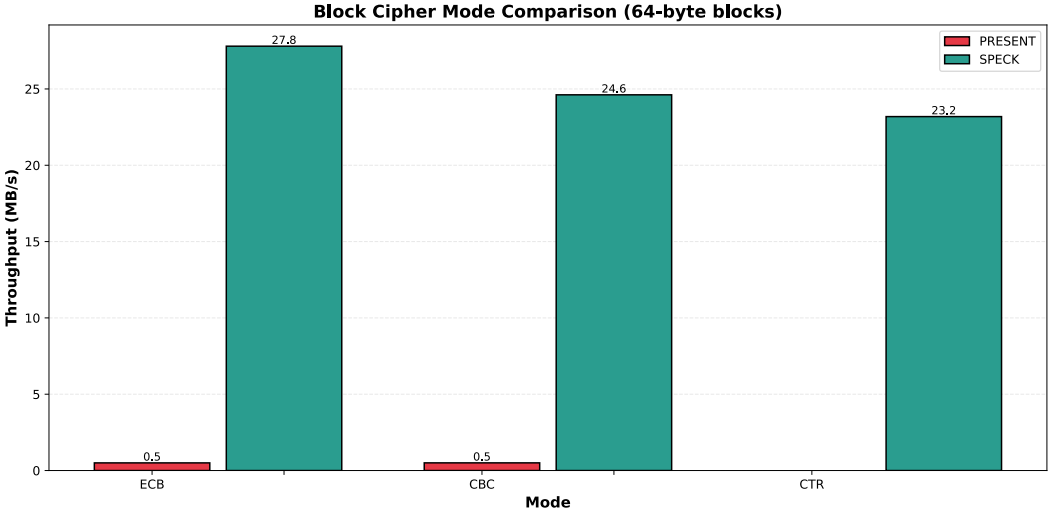
Primary benchmarks were conducted on Raspberry Pi Pico (ARM Cortex-M0+, 133 MHz).

Algorithm	ns/block	CPB	Mbps
PRESENT-80	23,665	2,958	0.34
PRESENT-128	24,542	3,068	0.33
SPECK-64/96	336	42	23.81
SPECK-64/128	338	42	23.67
ASCON-128	2,117	132	7.56

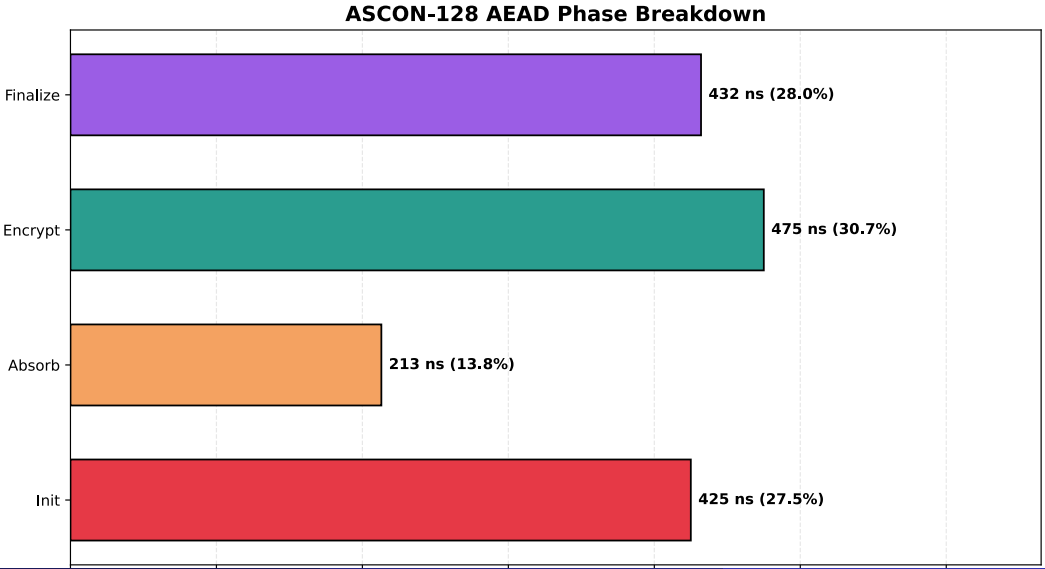
Visual Results: Block Cipher Comparison



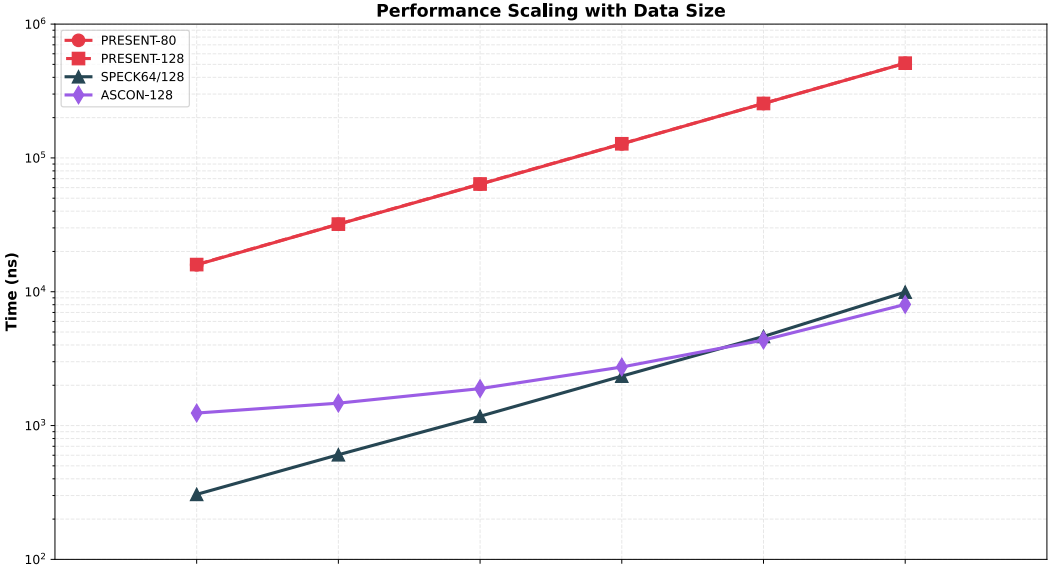
Visual Results: Mode Comparison



Visual Results: ASCON Phase Breakdown



Visual Results: Scaling Analysis



Analysis

Performance Observations

- **SPECK** is the fastest: 23.67 Mbps on Pico, approximately 70x faster than PRESENT
- **ASCON** provides good balance with built-in authentication (7.56 Mbps on Pico)
- **PRESENT** is the slowest on software but designed for hardware efficiency
- CTR mode offers best performance among encryption modes for SPECK
- ASCON's initialization overhead is amortized over larger message sizes

Memory Analysis

- **SPECK** has the smallest ROM footprint due to ARX-only operations (no lookup tables)
- **PRESENT** requires additional storage for S-box tables but remains compact
- **ASCON** has the largest code size due to 320-bit state and complex permutations

Conclusion

This study investigates lightweight cryptographic algorithms for resource-constrained embedded platforms.

PRESENT, SPECK, and ASCON were evaluated using Rust-based implementations on Raspberry Pi Pico.

Results show that SPECK provides superior performance, PRESENT offers minimal memory footprint, and ASCON delivers strong modern security with functional completeness.

Future Work

Short-Term

- **NIST KAT Tests**: Integrate official Known Answer Test vectors
- PRESENT: ISO/IEC 29192-2 test vectors
- SPECK: NSA-designed test vectors
- ASCON: NIST LWC standardization test vectors
- **Code Size Analysis**: Static analysis for .text/.data/.bss sections using cargo size
- **Statistical Rigor**: Report median/min/max across multiple benchmark runs instead of single run

Future Work

Medium-Term

- **FELICS Comparison**: Compare against established lightweight crypto benchmarks (felics.eu)
- **SUPERCOP-Style Timing**: Implement cycle-accurate benchmarking methodology
- **Side-Channel Discussion**: Analyze timing attack resistance for ARX ciphers (SPECK)

These directions will deepen the understanding of lightweight cryptographic design and facilitate secure IoT-system development.

Thank You